

Project Demo Planning

Date	23 June 2026
Team ID	SWTID-2026-1377
Project Name	Voice Based Concept Understanding Analyse
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demonstration showcases the **Voice-Based Concept Understanding Analyser** by explaining the problem, presenting the AI-powered solution, demonstrating voice-based concept evaluation, and highlighting the underlying NLP architecture and technologies. The demo concludes with a question-and-answer session to address evaluator queries.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Problem Statement	Explain the limitations of traditional assessments, the need for evaluating conceptual understanding, and the motivation behind developing the Voice-Based Concept Understanding Analyser.	3	Sindhura Devi
2	Solution Overview	Introduce the application's workflow, including voice input, speech-to-text conversion, NLP-based concept analysis, mastery score generation, and personalized feedback.	3	Sindhura Devi
3	System Architecture & Technology Walkthrough	Present the system architecture, explain the technology stack (Flask, Web Speech API, NLTK, Scikit-learn, JSON Knowledge Base), and describe the NLP processing pipeline	4	Sindhura Devi
4	Live Feature Demonstration	Demonstrate the application by opening the Home page, About page, recording a voice explanation (or entering text), submitting it for analysis, and displaying the mastery score, misconceptions, concept coverage, and personalized feedback	5	Sindhura Devi
5	Deployment & Future Enhancements	Demonstrate the deployed application on Vercel, briefly discuss future enhancements such as multilingual support, user authentication, and learning progress tracking.	2	Sindhura Devi
6	Q&A Session	Answer questions from evaluators regarding project implementation, NLP techniques, deployment, and future scope.	3	Sindhura Devi

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Explain why traditional assessments fail to measure conceptual understanding and introduce the motivation for the project.
2	Solution Overview	Describe the complete workflow: Voice Input → Speech-to-Text → NLP Analysis → Mastery Score → Personalized Feedback.
3	Live Feature Demonstration	Demonstrate the deployed application by navigating through the Home page, About page, recording or entering a concept explanation, and showing the generated analysis report.
4	Architecture & Technology Overview	Briefly explain the application architecture, NLP engine, Flask backend, and technology stack.
5	Q&A Session	Respond to evaluator questions and discuss future improvements and scalability.